Developing simulation and statistical evaluation tools for a nonlinear mixed-effects model

Extending and validating the joint model in Leaspy

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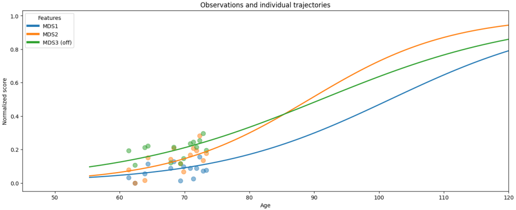
Section 1

Motivation and context

Longitudinal data: repeated observations over time

- The same patient i is observed at several chronological ages t.
- Visits can be sparse, irregular, unbalanced, and partially missing.
- Repeated measurements from one patient are correlated.



Dots: observations at successive visits. Curves: personalised trajectories.

We write $y_i(t)$ for the clinical scores observed for patient i at age t. **Nonlinear mixed-effects models** combine a population trajectory with patient-specific deviations.

Reconstruct disease dynamics from short, heterogeneous follow-ups.

Why simulation is indispensable here

Known truth

Real cohorts never reveal a patient's exact progression stage or patient-specific parameters.

Controlled tests

Check whether known parameters are recovered when visits are sparse or patients leave the study early.

Shareable cohorts

Support reproducibility and benchmarking without distributing patient records.

Internship gap. The joint-model paper used a separate research script; Leaspy's standard `simulate` interface supported only the logistic model.

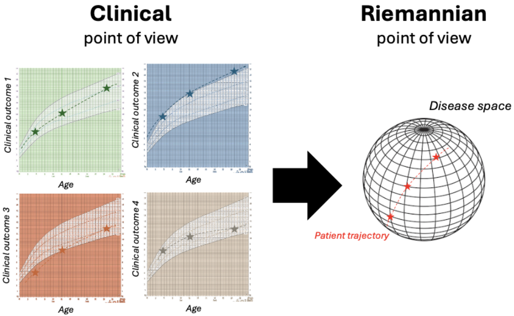
Goal: integrate joint simulation in Leaspy, then test the complete simulate–fit–personalise loop.

Simulation-study principles: Morris et al. [1].

Section 2

How Leaspy represents disease progression

From separate scores to a disease-space trajectory



- **Clinical view:** one curve per outcome against age.
- **Riemannian view:** their joint evolution is one path γ_i in disease space.

Leaspy represents several clinical scores through bounded logistic trajectories.

Figure: Ortholand [2]; disease course mapping: Schiratti et al. [3].

Nonlinear mixed effects: population and individual

Fixed effects: shared by the cohort
$(g, v_0, t_0, \sigma_\tau, \sigma_\xi)$

Random effects: one set per patient
$(\tau_i, \xi_i, \mathbf{w}_i, \zeta_i)$

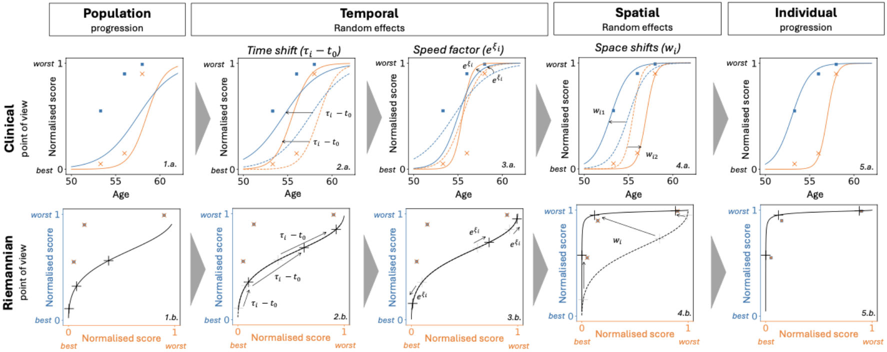
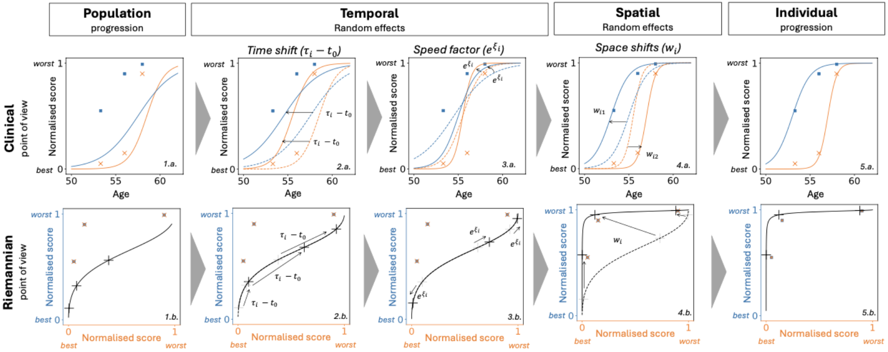


Figure adapted from Koval et al. [4].

Disease age separates disease stage from chronological age

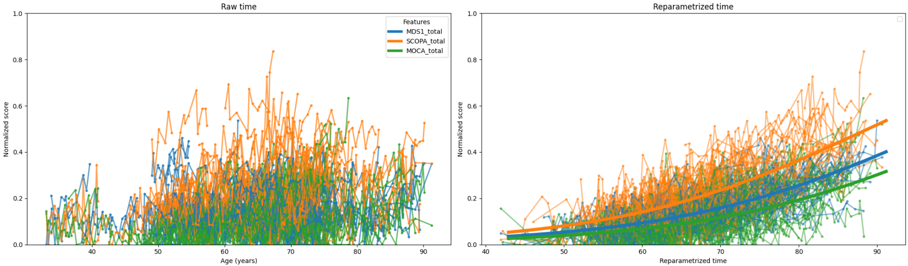


$$\boxed{\psi_i(t) = e^{\xi_i}(t - \tau_i) + t_0}$$

$\psi_i(t)$ is the latent disease age of patient i at chronological age t; τ_i is the individual reference age and ξ_i the log-acceleration.

Figure adapted from Koval et al. [4].

How random effects personalise the trajectory



τ_i
individual reference age; shifts the trajectory earlier or later

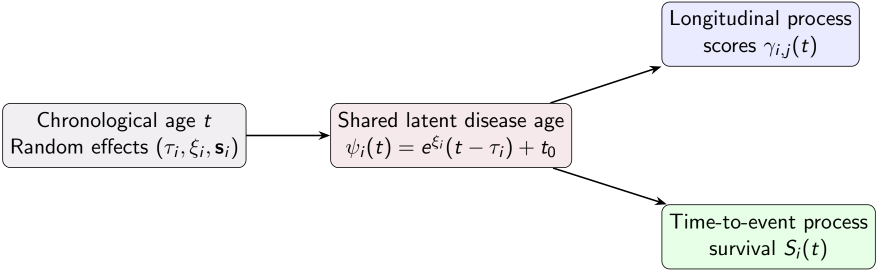
e^{ξ_i}
accelerates or slows disease progression

t_0
anchors the population disease timeline

Time shift, acceleration, and space shifts transform one population curve into an individual course.

Individual disease-course transformations: Koval et al. [4].

The joint model: one disease age, two processes



- **Competing events:** several mutually exclusive outcomes can end follow-up; observing one prevents the others from being observed for that patient.
- **Informative dropout:** follow-up ends for a reason related to the patient's unobserved health state, so the missing data are not independent of disease progression.

Joint latent-disease-age models: Ortholand et al. [5, 6].

Section 3

How the cohorts are simulated with the joint model

Two input layers must remain distinct

Learned by `fit`

- longitudinal shape: $t_0, \mathbf{g}, \mathbf{v}_0$
- heterogeneity: σ_τ, σ_ξ
- observation noise
- survival: $\boldsymbol{\nu}, \boldsymbol{\rho}$
- source mixing: $\mathbf{A}, \zeta$

Specified by the user

- cohort size N
- first-visit offset: $\bar{\delta}_f, \sigma_{\delta_f}$
- follow-up: $\overline{T}_f, \sigma_{T_f}$
- inter-visit gap: $\bar{\delta}_v, \sigma_{\delta_v}$
- minimum visit spacing
- or an explicit DataFrame visit schedule

The model determines what disease progression looks like; the user determines what the study looks like.

Simulation algorithm (1/2): create each follow-up

For each patient $i = 1, \ldots, N$:

1. Random effects

draw patient-specific effects

$$\xi_i \sim \mathcal{N}(0, \sigma_\xi^2)$$

$$\tau_i \sim \mathcal{N}(t_0, \sigma_\tau^2)$$

plus latent sources, when present

2. First visit

draw the first-visit time

$$t_{i,0} = \tau_i + \delta_{f_i}$$

$$\delta_{f_i} \sim \mathcal{N}(\bar{\delta}_f, \sigma_{\delta_f}^2)$$

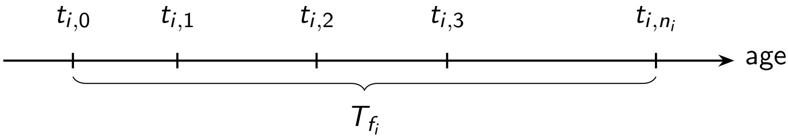
3. Visit schedule

draw inter-visit gaps

$$\delta_{v_{i,r}} \sim \mathcal{N}(\bar{\delta}_v, \sigma_{\delta_v}^2)$$

until the follow-up duration

$$T_{f_i} \sim \mathcal{N}(\overline{T}_f, \sigma_{T_f}^2)$$



Steps 2–3 are governed by the user-defined study design.

Simulation algorithm (2/2): outcomes and events

For each patient $i = 1, \ldots, N$:

4. Outcomes

compute disease age and trajectory

$\psi_i(t_{i,r}),\ \gamma_i$

draw bounded observations with Beta noise

5. Event time

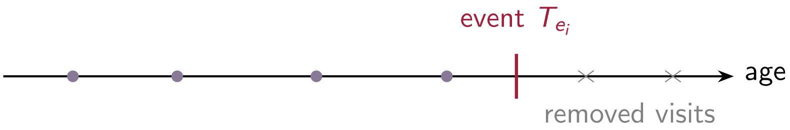
inverse-transform sampling from survival

$T_{e_i} \sim e^{-\xi_i}\mathcal{W}(\nu, \rho) + \tau_i$

6. Censoring

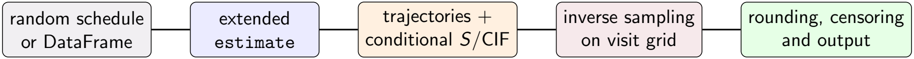
remove visits after T_{e_i}

right-censor events beyond the final visit



Output: longitudinal observations, an event or censoring time, and its indicator.

Integrating joint simulation inside Leaspy



1. Standard workflow

Extend `estimate` so joint models can use the same `simulate` workflow as other Leaspy models.

2. Competing events

Generalise event prediction and sampling from one event to several competing event types.

The other adaptations are mainly developer/API-oriented considerations and are detailed in the appendix.

Section 4

How the self-consistency of the simulation is tested

How simulation quality is assessed

Reference model
population truth θ^*

What is simulated exactly?

Reference: multivariate joint model fitted on the PRO-ACT ALS dataset
$J = 4$ longitudinal features, $K = 2$ latent sources, $L = 1$ event (death)

Bulbar function

Speech, salivation, and swallowing items.

Gross-motor function

Turning in bed; walking; climbing stairs.

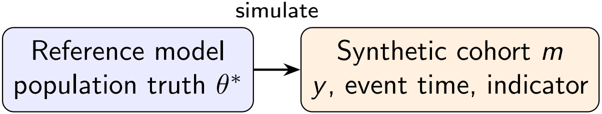
Fine-motor function

Handwriting; cutting food/handling utensils; dressing and hygiene.

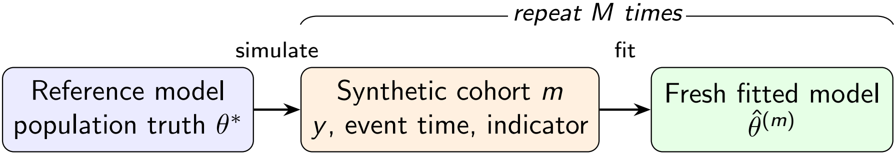
Total function

Overall ALSFRS-R: all items, including the preceding domains and respiratory items.

Step 1: simulate from a known reference model



Step 2: fit a fresh model to each synthetic cohort



Experimental setting: deliberately information-rich

Replications

$N = 1000$ patients
$M = 200$ cohorts

Visit process

first offset -2 ± 1 y
follow-up 6 ± 2 y
interval 0.083 ± 0.042 y
minimum 0.05 y

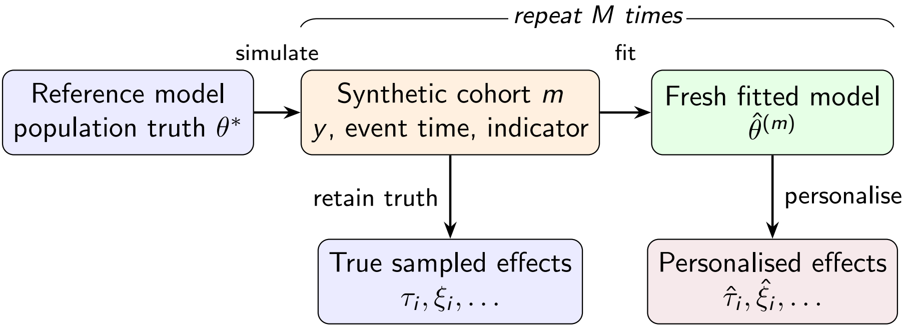
Estimation

MCMC-SAEM
70 000 fit iterations
5 000 personalisation iterations

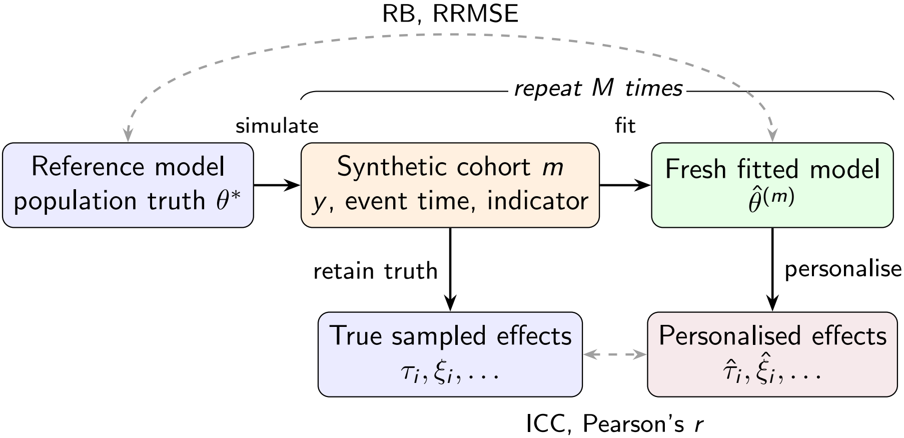
Interpretation boundary: short visit intervals and long follow-up create a favourable dense-observation setting.

MCMC-SAEM for nonlinear mixed-effects models: Kuhn et al. [7].

Step 3: retain and recover individual effects



Step 4: compare population and individual recovery



Four recovery metrics, two inference levels

Fixed effects / population parameters

Define the relative error in repetition m as

$$e_m = \frac{\hat{\theta}^{(m)} - \theta^*}{|\theta^*|} \times 100.$$

$$\mathrm{RB} = \frac{1}{M} \sum_{m=1}^{M} e_m$$

signed systematic error; best value: 0%.

$$\mathrm{RRMSE} = \sqrt{\frac{1}{M} \sum_{m=1}^{M} e_m^2}$$

total error from bias and dispersion; best value: 0%.

Individual random effects

ICC(3,1)

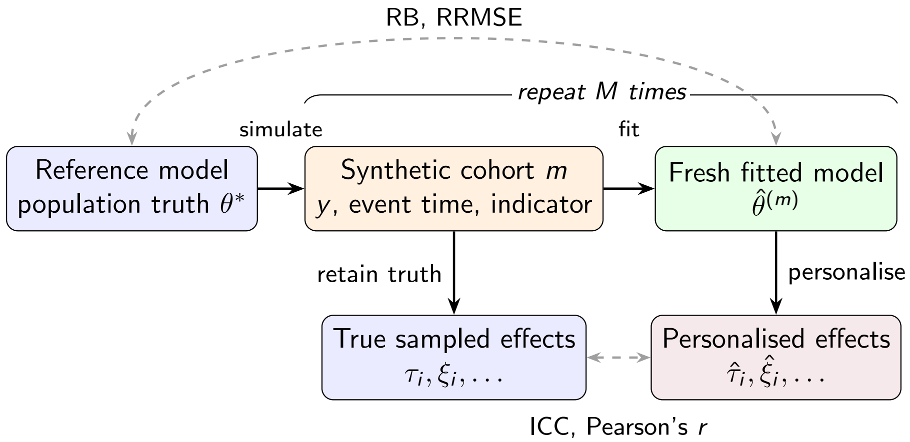
Consistency between true and personalised values while accounting for disagreement beyond rank ordering.

Pearson's r

Strength of linear association; invariant to location and scale changes.

Best value for both: 1.

The complete self-consistency experiment



Self-consistency asks whether the model can recover what it generated.

Section 5

Results of the experiments

Fixed effects / population parameters: recovery

Component	Parameters	Interpretation	RB (%)	RRMSE (%)
Temporal	$t_0, \sigma_\tau, \sigma_\xi$	reference and variability	0.03 to 0.44	0.83 to 2.32
Survival	ρ, ν	Weibull shape and scale	0.10 to 2.33	5.01 to 5.43
Noise	$\sigma^{(0:3)}$	observation noise	-4.60 to -1.26	1.47 to 4.73
Position	$g^{(0:3)}$	logistic position	-8.80 to -5.92	8.09 to 12.61
Velocity	$v_0^{(0:3)}$	logistic velocity	1.12 to 6.46	4.52 to 10.53

Best recovered: temporal population parameters, supported by many repeated observations per patient.

Hardest: trajectory positions g and bulbar velocity $v_0^{(0)}$, which can compensate for each other.

Individual random effects: recovery (1/2)

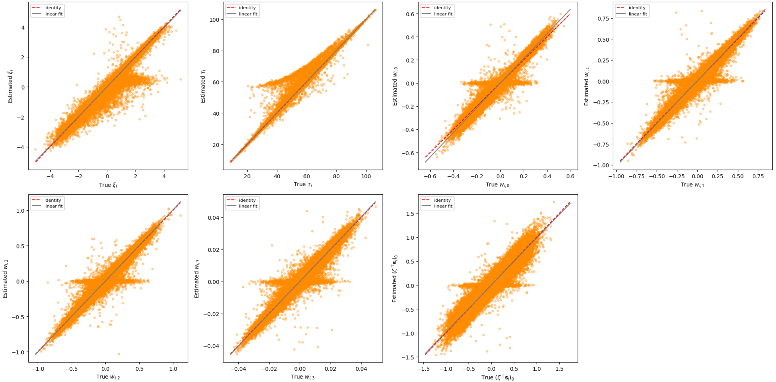
Effect	ICC(3,1)	Pearson's r
ξ_i	0.988 ± 0.004	0.989 ± 0.004
τ_i	0.995 ± 0.002	0.995 ± 0.002
$w_{i,0}$	0.987 ± 0.006	0.992 ± 0.003
$w_{i,1}$	0.989 ± 0.003	0.990 ± 0.003
$w_{i,2}$	0.990 ± 0.003	0.991 ± 0.003
$w_{i,3}$	0.987 ± 0.003	0.988 ± 0.003
$(\zeta^\top \mathbf{s}_i)_0$	0.965 ± 0.027	0.978 ± 0.013

Mean $\pm$ standard deviation across repetitions.

Strong agreement for every evaluated individual effect: all ICCs exceed 0.96.

The **survival shift** is least precise: it is informed by at most one possibly censored event per patient.

Source coordinates $\mathbf{s}_i$ are rotation-dependent, so invariant feature shifts $\mathbf{w}_i = \mathbf{A}\mathbf{s}_i$ and survival shifts are compared instead.



Aggregate ICC is high, but some low-information patients are pulled toward the population mean.

True sampled values (x-axis) vs personalised estimates (y-axis). Dashed red: identity; grey: linear fit. Points are pooled over patients and repetitions.

Section 6

Conclusions

Take-home messages

Goal: integrate joint simulation in Leaspy, then test the complete simulate–fit–personalise loop.

1. **Integration achieved:** joint simulation now uses the standard Leaspy API, including survival, censoring, and competing events.
2. **Validation achieved:** the complete loop is self-consistent in the favourable dense setting studied.
3. **Main result:** temporal and individual effects are recovered very well; trajectory positions and survival-related quantities are less precisely identified.
4. **Scope:** recovery under sparse visits, heavier censoring, or model misspecification remains to be evaluated.

Ethical and sustainability considerations CS ONLY

Patient privacy

Synthetic cohorts do not correspond to real patients and can facilitate sharing.

They still require clear labelling, provenance, governance, and disclosure-risk assessment.

Computational cost

200 fits $\times$ 70 000 iterations plus personalisation is not negligible.

Small preliminary runs and saved intermediates avoid unnecessary recomputation.

Software responsibility

Errors can propagate into scientific conclusions.

Unit tests, functional tests, documentation, and peer review are part of methodological rigour.

Synthetic does not mean risk-free, and reproducible does not mean cost-free.

References I

[1] Tim P Morris, Ian R White, and Michael J Crowther. "Using simulation studies to evaluate statistical methods". In: *Statistics in medicine* 38.11 (2019), pp. 2074–2102.

[2] Juliette Ortholand. "Joint modelling of events and repeated observations: an application to the progression of Amyotrophic Lateral Sclerosis". PhD thesis. Sorbonne Université, 2024.

[3] Jean-Baptiste Schiratti et al. "A Bayesian mixed-effects model to learn trajectories of changes from repeated manifold-valued observations". In: *Journal of Machine Learning Research* 18.133 (2017), pp. 1–33.

[4] Igor Koval et al. "AD Course Map charts Alzheimer's disease progression". In: *Scientific Reports* 11.1 (2021), p. 8020.

[5] Juliette Ortholand, Stanley Durrleman, and Sophie Tezenas du Montcel. "A joint spatiotemporal model for multiple longitudinal markers and competing events". In: *arXiv preprint arXiv:2501.08960* (2025).

[6] Juliette Ortholand et al. "Joint model with latent disease age: Overcoming the need for reference time". In: *Statistical Methods in Medical Research* 35.2 (2026), pp. 225–239.

References II

[7] Estelle Kuhn and Marc Lavielle. "Maximum likelihood estimation in nonlinear mixed effects models". In: *Computational statistics & data analysis* 49.4 (2005), pp. 1020–1038.

Appendix: notation and normalised disease space

$i = 1, \ldots, N$	patient
$r = 1, \ldots, n_i$	visit
$j = 1, \ldots, J$	longitudinal feature
$k = 1, \ldots, K$	latent source
$l = 1, \ldots, L$	event type

- $t_{i,r,j}$: chronological age
- $y_{i,r,j}$: observed score
- $\gamma_{i,j}$: latent individual trajectory
- $\epsilon_{i,r,j}$: observation noise

With J normalised clinical scores, the disease space is $(0,1)^J$. The chosen metric produces bounded logistic trajectories in this space.

$$y_{i,r,j} = \gamma_{i,j}(t_{i,r,j}) + \epsilon_{i,r,j}$$

Appendix: full longitudinal model

$$\begin{cases} \psi_i(t) = e^{\xi_i}(t - \tau_i) + t_0, \\ \mathbf{w}_i = \mathbf{A}\mathbf{s}_i, \\ \gamma_{i,j}(t) = \left[1 + g_j \exp\left(-\frac{(1+g_j)^2}{g_j}\left(v_{0,j}(\psi_i(t) - t_0) + w_{i,j}\right)\right)\right]^{-1}, \\ y_{i,r,j} = \gamma_{i,j}(t_{i,r,j}) + \epsilon_{i,r,j}, \qquad \epsilon_{i,r,j} \sim \mathcal{N}(0, \sigma_j^2). \end{cases}$$

- $p_j = 1/(1 + g_j)$ is the population trajectory value at t_0.
- $v_{0,j}$ controls the velocity at the reference time.
- $w_{i,j}$ changes the relative ordering of feature trajectories.

Appendix: survival and latent-source coupling

Single-event survival on disease age

$$S_i(t) = \mathbf{1}_{\psi_i(t)>t_0} \exp\left[-\left(\frac{\psi_i(t)-t_0}{\nu}\right)^{\rho}\right] + \mathbf{1}_{\psi_i(t)\le t_0}.$$

Event-specific scale with sources

For event l,

$$\nu_{i,l} = \nu_l \exp\left[-\xi_i - \frac{1}{\rho_l}(\boldsymbol{\zeta}^\top \mathbf{s}_i)_l\right].$$

- Each competing event has its own Weibull pair (ν_l, ρ_l).
- $\mathrm{CIF}_{i,l}(t)$ gives the probability that cause l has occurred by t in the presence of competing causes.

Appendix: discrete event sampling in Leaspy

$$F_i^{\mathrm{cond}}(t_{i,r}) = \begin{cases} 1 - S_i(t_{i,r})/S_i(t_{i,0}), & \text{single event,} \\ \sum_{l=1}^{L} \mathrm{CIF}_{i,l}(t_{i,r})/S_i(t_{i,0}), & \text{competing events.} \end{cases}$$

1. Draw $U \sim \mathcal{U}[0,1]$.
2. Select the first planned visit where $F_i^{\mathrm{cond}}(t_{i,r}) \geq U$.
3. If no visit crosses U, right-censor at the last visit.
4. For competing risks, sample the cause from incremental cause-specific CIFs.

This ensures internal API reuse, but discretises event times onto the visit grid.

Appendix: additional Leaspy adaptations

Visit schedules and study parameters

The simulator accepts either a random visit process or a user-provided schedule. Missing random-mode study parameters can be estimated from an input dataset.

Bounded observation noise

Longitudinal observations use Beta noise. Near score boundaries, an infeasible variance is clamped to $0.99\,\mu(1-\mu)$ and a warning identifies the affected observation.

Competing causes and rounding

Event types are sampled from incremental cause-specific cumulative incidences. After visit-time rounding, a final censoring pass removes any visit moved beyond the event time.

Appendix: detailed population recovery (1/2)

Parameter	θ^*	$\bar{\hat{\theta}}$	RB (%) [95% CI]	RRMSE (%) [95% CI]	RSE [95% CI]
$g^{(0)}$	10.461	9.595	-8.27 [$-9.41, -7.23$]	11.41 [10.60, 12.28]	0.086 [0.076, 0.095]
$g^{(1)}$	2.326	2.122	-8.80 [$-10.03, -7.52$]	12.61 [11.55, 13.66]	0.099 [0.087, 0.111]
$g^{(2)}$	1.951	1.803	-7.60 [$-8.71, -6.52$]	11.16 [10.26, 12.06]	0.089 [0.079, 0.098]
$g^{(3)}$	3.415	3.213	-5.92 [$-6.68, -5.17$]	8.09 [7.46, 8.76]	0.059 [0.052, 0.066]
$v_0^{(0)}$	0.078	0.083	6.46 [5.32, 7.65]	10.53 [9.56, 11.53]	0.078 [0.070, 0.086]
$v_0^{(1)}$	0.257	0.263	2.23 [1.54, 2.93]	5.57 [5.05, 6.12]	0.050 [0.045, 0.054]
$v_0^{(2)}$	0.238	0.241	1.12 [0.52, 1.74]	4.52 [4.12, 4.92]	0.043 [0.039, 0.047]
$v_0^{(3)}$	0.133	0.135	2.05 [1.37, 2.73]	5.25 [4.75, 5.78]	0.048 [0.043, 0.052]

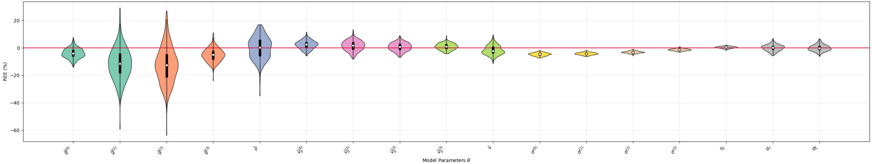
Intervals are 95% percentile bootstrap confidence intervals from 2,000 resamples of the $M = 200$ cohort-level estimates.

Appendix: detailed population recovery (2/2)

Parameter	θ^*	$\bar{\hat{\theta}}$	RB (%) [95% CI]	RRMSE (%) [95% CI]	RSE [95% CI]
ρ	1.849	1.851	0.10 $[-0.61, 0.81]$	5.01 $[4.50, 5.51]$	0.050 $[0.045, 0.055]$
ν	4.047	4.142	2.33 $[1.64, 3.00]$	5.43 $[4.97, 5.91]$	0.048 $[0.043, 0.053]$
$\sigma^{(0)}$	0.069	0.065	-4.60 $[-4.76, -4.45]$	4.73 $[4.58, 4.89]$	0.012 $[0.011, 0.013]$
$\sigma^{(1)}$	0.079	0.075	-4.16 $[-4.28, -4.04]$	4.25 $[4.14, 4.38]$	0.009 $[0.009, 0.010]$
$\sigma^{(2)}$	0.074	0.072	-3.08 $[-3.17, -2.97]$	3.17 $[3.07, 3.27]$	0.008 $[0.007, 0.009]$
$\sigma^{(3)}$	0.045	0.044	-1.26 $[-1.37, -1.15]$	1.47 $[1.37, 1.56]$	0.008 $[0.007, 0.009]$
t_0	56.699	56.948	0.44 $[0.35, 0.53]$	0.83 $[0.76, 0.90]$	0.007 $[0.006, 0.008]$
σ_τ	11.630	11.639	0.08 $[-0.24, 0.38]$	2.32 $[2.10, 2.54]$	0.023 $[0.021, 0.025]$
σ_ξ	1.078	1.079	0.03 $[-0.27, 0.37]$	2.32 $[2.11, 2.56]$	0.023 $[0.021, 0.026]$

θ^* is the generating value; $\bar{\hat{\theta}}$ is the mean estimate across the 200 independently fitted cohorts.

Appendix: population relative-error distributions



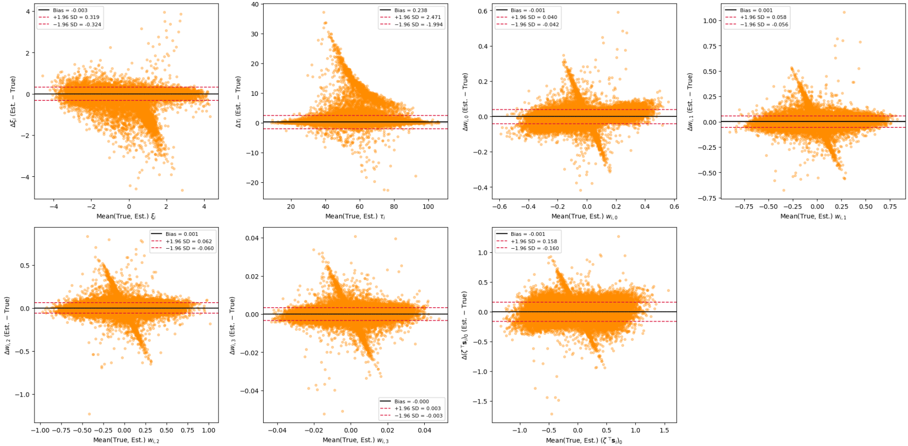
Red: zero relative error. Temporal parameters concentrate closest to zero; trajectory positions $g^{(j)}$ show the largest systematic underestimation.

Appendix: detailed individual-effect recovery

Effect	ICC(3,1)	Pearson's r
ξ_i	0.988 ± 0.004	0.989 ± 0.004
τ_i	0.995 ± 0.002	0.995 ± 0.002
$w_{i,0}$	0.987 ± 0.006	0.992 ± 0.003
$w_{i,1}$	0.989 ± 0.003	0.990 ± 0.003
$w_{i,2}$	0.990 ± 0.003	0.991 ± 0.003
$w_{i,3}$	0.987 ± 0.003	0.988 ± 0.003
$(\zeta^\top \mathbf{s}_i)_0$	0.965 ± 0.027	0.978 ± 0.013

Values are mean $\pm$ standard deviation across the $M = 200$ cohorts, as reported in the report.

Appendix: Bland–Altman diagnostic



Black: mean difference; red: limits of agreement. Biases are close to zero for all effects.

Appendix: software-level validation

Simulation unit tests

- non-empty random-mode output
- expected patient count
- strictly increasing visit times
- DataFrame schedules are respected
- all model features are returned

`estimate` **functional tests**

- univariate numerical correctness
- raw/DataFrame consistency
- MultiIndex preservation
- multivariate output shape: $n_{\mathrm{timepoints}} \times (J + L)$

Appendix: limitations and next experiments

What the reported experiment establishes

Self-consistency of the integrated pipeline for $N = 1000$ under a favourable, dense visit design.

What it does not establish

Universal recovery under sparse visits, heavier censoring, smaller cohorts, or model misspecification.

Perspectives stated in the report

Vary sample size, censoring intensity, visit sparsity, and misspecification; explore more flexible visit schedules and noise models; align continuous-time event generation more closely with the grid-based implementation.